Systems Analyst Assignments

Systems Analysis and Design | Given System: Hospital Management System (HMS)

Assignment 1: Skills of a Systems Analyst

A Hospital Management System manages patient registration, appointments, billing, pharmacy, and lab reports in one place. A systems analyst needs four main skills to build it successfully.

1. Interpersonal Skills

These help the analyst communicate with different people. In an HMS project, the analyst must talk to doctors, nurses, receptionists, pharmacists, and administrators, who all have different needs and technical knowledge. Good listening, patience, and clear explanation help the analyst understand their problems. Teamwork and negotiation help resolve conflicts, such as when doctors and the billing department want different things.

2. Technical Skills

Technical skills let the analyst understand how the system will be built. For an HMS, the analyst should know database design, programming basics, networking, and system security. Since medical data is sensitive, data protection and access control are especially important. These skills help the analyst work with developers and choose tools such as MySQL, Java, or web technologies.

3. Analytical Skills

Analytical skills mean the ability to study a problem, break it into parts, and find the best solution. The analyst studies the current manual process, such as paper files and long queues, and finds problems like duplicate records, delays, and billing errors. Tools like DFDs, flowcharts, and use case diagrams help model the process and show how the new system can fix these problems.

4. Management Skills

Management skills help the analyst plan and control the project. The analyst must manage time, budget, and resources, and divide tasks among team members. Risk management is also important, for example handling data loss when moving from paper to digital records. Good management ensures the project finishes on time and the hospital staff accept the system.

Assignment 2: Methodologies for Requirement Determination

Requirement determination means finding out what users and the organization need from the new system. If requirements are collected wrongly, the system will fail. The following methodologies follow the six steps shown in the slide.

Step 1: Stakeholder Identification

We identify everyone affected by the HMS: patients, doctors, nurses, receptionists, lab staff, pharmacists, accountants, and hospital management, so that no one is missed.

Step 2: Requirement Elicitation Techniques

This is where requirements are actually collected. Common techniques are:

- **Interviews:** Detailed and accurate, but time-consuming.
- **Questionnaires:** Quick for many nurses and staff, but answers may be shallow.
- **Observation:** Shows how registration and billing really work, but people may behave differently when watched.
- **Document Analysis:** Studying old patient files, bills, and report formats.
- **Prototyping:** A sample appointment screen gives early user feedback.
- **JAD (Joint Application Development):** Users and developers decide requirements together in workshops.

Step 3: Requirement Documentation

Requirements are written in an SRS (Software Requirements Specification), divided into functional requirements (e.g., book appointments, generate bills) and non-functional requirements (e.g., security, fast response, 24/7 availability).

Step 4: Requirements Analysis and Prioritization

Requirements are checked for conflicts and duplication, then ranked using a method like MoSCoW. Patient record management is a "must have," while an SMS reminder may be a "could have."

Step 5: Verification and Validation

The team checks that requirements are complete and correct (verification) and match user needs (validation), using reviews, walkthroughs, and prototype demos.

Step 6: Requirements Management and Change Control

Requirements often change, such as a new health rule or a new department. A change control process records, reviews, and approves each change so the project stays under control.

Conclusion

For the HMS, no single technique is enough. Combining interviews, observation, questionnaires, and prototyping, and following all six steps, helps the analyst collect clear and correct requirements.